Blockchain in Bilateral Derivatives Markets: Quantifying the Impact on XVA Pricing, Counterparty Credit Risk, Collateral, and Regulatory Capital

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Abstract

The ISDA Standard Initial Margin Model (SIMM) calibrates risk weights to the 99th percentile of N-day market moves, where N is the Margin Period of Risk (MPoR). The BCBS-IOSCO framework mandates a minimum MPoR of ten business days for bilateral non-cleared derivatives—deliberately set twice the five-day floor for centrally cleared instruments as a structural clearing incentive. Every risk weight in ISDA SIMM v2.8+2506 across all six risk classes embeds this assumption.

This paper provides a comprehensive quantitative analysis of how blockchain trade confirmation and atomic settlement restructure the bilateral derivatives cost stack across four dimensions: XVA pricing, counterparty credit risk, collateral economics, and SA-CCR regulatory capital. We identify eight mechanisms through which on-chain infrastructure reduces costs and provide closed-form quantification of each. The primary mechanism—regulatory recognition of a compressed MPoR for blockchain-confirmed trades—reduces every SIMM risk weight by $\sqrt{5/10} = 1/\sqrt{2} \approx 0.7071$, implying a portfolio-invariant 29.3% reduction in initial margin across all risk classes simultaneously.

On XVA pricing: CVA falls 60–70% as expected exposure profiles collapse under continuous margining; FVA falls 80–100% as automated variation margin eliminates the uncollateralised funding gap; MVA falls 40–55% through lower IM balance and reduced net funding spread on tokenised collateral; KVA falls 68–84% via SA-CCR maturity factor compression. All-in XVA falls 60–66%. On counterparty credit risk: PFE and EPE profiles collapse as MPoR compresses the close-out horizon; credit limit capacity increases 3.6–7.2 times for typical bilateral portfolios. On collateral: the SIMM IM balance falls 29–40% and tokenised IM earns 45–50 bp of additional yield while posted, directly reducing MVA. On regulatory capital: SA-CCR EAD falls 68–84%, producing equivalent savings in CCR capital requirements.

We further demonstrate that the regulatory logic underlying the bilateral MPoR penalty inverts under blockchain settlement, raising the question of whether blockchain-confirmed bilateral derivatives

should warrant lower capital requirements than centrally cleared instruments. We also address implications for buy-side institutions, corporate hedgers, and financial regulators. CFTC rulemaking targeted for August 2026 provides the legislative vehicle.

Keywords: ISDA SIMM, XVA, CVA, FVA, MVA, KVA, PFE, EPE, MPoR, Blockchain, Atomic Settlement, Tokenised Collateral, SA-CCR, Counterparty Credit Risk, CFTC, BCBS-IOSCO

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Author's Note on AI Assistance. The intellectual framework, analytical thesis, and industry insights in this paper are entirely the author's own, developed through fifteen years of practitioner experience in derivatives valuation, counterparty credit risk, and XVA. The central argument—that blockchain settlement compresses the Margin Period of Risk and restructures the bilateral derivatives cost stack—originated with the author and does not appear in prior literature to the author's knowledge. Large language model AI was used to assist with mathematical formalisation and LaTeX typesetting of equations, prose drafting under the author's direction, and structural organisation of the paper. All analytical judgements, numerical assumptions, regulatory interpretations, and conclusions were made by the author. The AI served as a sophisticated drafting and formalisation tool; the intellectual contribution is the author's. The author believes transparent disclosure of AI assistance in academic and practitioner work is both ethically correct and professionally important as these tools become standard in research practice.

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1. Introduction

When Wall Street's paperwork crisis of 1968 threatened to paralyse equity markets, the Depository Trust Company resolved it not by improving trading, but by dematerialising the infrastructure beneath it. Physical certificates became electronic book entries. $87 trillion in equities now clears annually on that foundation. Fifty years later, bilateral derivatives markets face an equivalent structural failure—not in volume, but in architecture. ISDA confirmations negotiated over hours. Initial margin disputed between independently computed sensitivity inputs, generating daily reconciliation workflows that consume billions in operational cost annually. Collateral settling T+1 across correspondent chains. Underlying all of it, a regulatory assumption—the ten-day *Margin Period of Risk (MPoR)*—set in 2013 not because it reflected close-out reality, but because regulators wished to make bilateral derivatives structurally more expensive than cleared equivalents (BCBS-IOSCO, 2015).

The ten-day bilateral MPoR is embedded in every risk weight in ISDA SIMM v2.8+2506. Since SIMM calibrates to the 99th percentile of N-day market moves, and market move distributions scale with $\sqrt{N}$, any change in the regulatory MPoR produces an immediate, portfolio-invariant, proportional change in every risk weight across every risk class simultaneously. The mathematics is not in dispute. The question is whether regulators will recognise a shorter MPoR for bilateral derivatives whose settlement mechanics are demonstrably equivalent to—or superior to—centrally cleared instruments.

This paper argues they must, and provides the quantitative case across four distinct dimensions. First, XVA pricing: CVA, FVA, MVA, and KVA each compress through specific channels linked to MPoR compression and continuous margining. Second, counterparty credit risk: PFE and EPE profiles collapse as the close-out horizon shrinks and intraday exposure accumulation is eliminated. Third, collateral economics: initial margin falls 29–40%, and tokenised IM earns yield while posted, reducing the cost of funding collateral. Fourth, regulatory capital: SA-CCR EAD falls by the same MPoR scaling factor as SIMM, producing identical proportional savings in CCR capital requirements.

The paper proceeds as follows. Section 2 reviews the ISDA SIMM framework and MPoR scaling law. Section 3 identifies eight blockchain mechanisms. Section 4 traces the XVA impact. Section 5 addresses counterparty credit risk: PFE, EPE, and exposure profile dynamics. Section 6 treats collateral economics. Section 7 covers SA-CCR and regulatory capital. Section 8 addresses implications for buy-side institutions, corporate hedgers, and regulators. Section 9 sets out the regulatory roadmap. Section 10 concludes.

2. ISDA SIMM, MPoR, and the Bilateral Penalty

2.1 The Three-Level SIMM Aggregation

ISDA SIMM v2.8+2506 is a sensitivity-based parametric model computing initial margin from portfolio Greeks across six risk classes: GIRR, CSR-Qualifying, CSR-Non-Qualifying, FX, Equity, and Commodity. Aggregation proceeds in three levels.

Level 1: Weighted Sensitivities

For each risk factor k within bucket b:

$$\mathrm{WS}_k \;=\; \mathrm{RW}_k \;\times\; s_k \;\times\; \mathrm{CR}_k \tag{2.1}$$

where RW_k is the supervisory risk weight (the 99th percentile of the N-day move distribution); s_k is the net sensitivity (delta, vega, or curvature); and the concentration risk factor is:

$$\mathrm{CR}_k \;=\; \max\left(1,\; \sqrt{\frac{\left|\sum_{j\in b} s_j\right|}{T_b}}\right) \tag{2.2}$$

with T_b the supervisory concentration threshold for bucket b.

Level 2: Bucket Aggregation

$$K_b \;=\; \sqrt{\max\left(0,\; \sum_{k\in b} \mathrm{WS}_k^2 \;+\; \sum_{k\in b}\sum_{\ell\neq k} \rho_{k\ell}\,\mathrm{WS}_k\,\mathrm{WS}_\ell\right)} \tag{2.3}$$

where $\rho_{k\ell} \in [-1, 1]$ is the supervisory intra-bucket correlation. The $\max(0, \cdot)$ prevents negative bucket-level IM for strongly negatively correlated positions.

Level 3: Risk Class and Portfolio Aggregation

$$\mathrm{IM}_{\mathrm{rc}} \;=\; \sqrt{\sum_b K_b^2 \;+\; \sum_b \sum_{c\neq b} \gamma_{bc}\, S_b\, S_c} \tag{2.4}$$

where $S_b = \max(-K_b,\, \min(K_b,\, \sum_{k\in b} \mathrm{WS}_k))$ is the clipped net sensitivity at the bucket level, and γ_{bc} the inter-bucket correlation. Total SIMM for a product class:

$$\mathrm{SIMM}_{\mathrm{pc}} \;=\; \sqrt{\sum_{\mathrm{rc}} \mathrm{IM}_{\mathrm{rc}}^2} \tag{2.5}$$

SIMM does not recognise diversification across product classes; total IM sums SIMM_{pc} across all product classes.

2.2 The MPoR Calibration and the Fundamental Scaling Law

SIMM risk weights are calibrated as the 99th percentile of overlapping N-day risk factor returns, where $N = \text{MPoR}$. Under the diffusion framework embedded in the calibration—consistent with the FRTB-SA approach on which SIMM is modelled—risk factor increments are stationary with variance proportional to horizon:

$$\text{RW}(N) \;=\; \text{RW}(1) \;\times\; \sqrt{N} \tag{2.6}$$

Consequently, for any change in regulatory MPoR from N_1 to N_2 days:

$$\frac{\text{RW}(N_2)}{\text{RW}(N_1)} \;=\; \sqrt{\frac{N_2}{N_1}} \tag{2.7}$$

This holds simultaneously and identically across all risk weights in all six risk classes. Because SIMM IM is positively homogeneous in the weighted sensitivities (equations 2.1–2.5), a uniform scaling of all risk weights by λ scales total SIMM IM by λ.[1]

Proposition 2.1 (Portfolio-Invariant MPoR Scaling of SIMM)**.** *Under the positive homogeneity assumption, reducing the regulatory MPoR from N_1 to N_2 business days reduces total SIMM IM by the factor $\sqrt{N_2/N_1}$ simultaneously across all risk classes, all netting sets, and all portfolio compositions.*

For $N_1 = 10$, $N_2 = 5$:

$$\frac{\text{SIMM}_{\text{blockchain}}}{\text{SIMM}_{\text{traditional}}} \;=\; \sqrt{\frac{5}{10}} \;=\; \frac{1}{\sqrt{2}} \;\approx\; 0.7071 \tag{2.8}$$

Corollary 2.1.1. *Regulatory recognition of a five-day MPoR for blockchain-confirmed bilateral trades produces a 29.3% reduction in total SIMM IM that is invariant to portfolio composition, asset class mix, counterparty credit quality, and netting set size.*

2.3 The MPoR Doubling Mechanism Under Dispute Conditions

Under BCBS-IOSCO (2015), paragraph 3.4, the minimum MPoR doubles to **twenty business days** when a netting set has experienced two or more unresolved margin disputes

[1] Exact when $\text{CR}_k = 1$ for all k. With concentration risk, CR_k depends nonlinearly on s_k through equation (2.2), breaking exact homogeneity. For portfolios with moderate concentration the approximation is negligible; for highly concentrated books the true reduction will be slightly less than $\sqrt{N_2/N_1}$.

within the preceding two quarters. The consequence:

$$\frac{\mathrm{RW}(20)}{\mathrm{RW}(10)} = \sqrt{\frac{20}{10}} = \sqrt{2} \approx 1.414 \tag{2.9}$$

A 41.4% increase in all risk weights for the duration of the dispute. Section 3.1 demonstrates that the primary cause of bilateral SIMM disputes is categorically eliminated by blockchain.

3. Eight Mechanisms of Blockchain-Driven Cost Reduction

3.1 Mechanism 1: Dispute Elimination and Prevention of MPoR Doubling

Under bilateral SIMM, each counterparty independently computes CRIF sensitivities using its proprietary pricing model. A ten-year USD IRS produces materially different IR01 values depending on curve construction and SOFR discount conventions. Both are internally self-consistent; neither is definitively wrong. But they generate divergent WS_k, different K_b, and different SIMM amounts—triggering reconciliation workflows and, when unresolved, the twenty-day doubling of equation (2.9).

With on-chain trade confirmation and a shared pricing oracle, both parties derive CRIF sensitivities from the same immutable on-chain record. The hash of the agreed sensitivity vector $\boldsymbol{s} = (s_1, \ldots, s_K)$ is anchored at the confirmation transaction. Disputes are *structurally impossible*. The twenty-day MPoR doubling condition cannot trigger. Excess collateral held pending dispute resolution—estimated at 5–15% of SIMM IM—is released immediately. Operational cost of reconciliation falls by 85–90%.

3.2 Mechanism 2: On-Chain CRIF and Identical Sensitivity Inputs

The shared on-chain CRIF eliminates the entire reconciliation workflow. The five-step process of compute, format, transmit, reconcile, recompute is replaced by a single read of the on-chain sensitivity record. Both parties compute SIMM from an identical source. No reconciliation platform is required.

3.3 Mechanism 3: Tokenised IM Earns Yield While Posted

Traditional cash IM earns the overnight index rate (SOFR $\approx$ 4.30%, March 2026). A firm funding at SOFR +65 bp bears a net funding cost of 65 bp on posted IM. Tokenised T-bills or money market fund shares earn approximately 4.75–4.80%—45–50 bp above SOFR. The IM poster retains this yield while collateral is held in smart contract escrow, reducing the effective funding spread $s_f(t)$ from approximately 65 bp to 15–20 bp, a 70–77% reduction in the MVA funding cost component. CFTC Letter 25-39 (December 2025) explicitly enables this structure (CFTC, 2025).

3.4 Mechanism 4: Cleaner Dynamic Backtesting Data

The dynamic backtest counts days where $|\Delta V_t| > \widehat{\mathrm{SIMM}}_1(t) = \mathrm{SIMM}_{10}(t)/\sqrt{10}$ over a 250-day rolling window. Under traditional bilateral margining, P&L during a dispute window is ambiguous—IM was not transferred on schedule, introducing systematic noise into the exception count. Under blockchain, every transfer is atomic, timestamped, and unambiguous. Every observation is clean. Spurious exceptions are eliminated, reducing unnecessary remediation IM add-ons.

3.5 Mechanism 5: Reduced Haircuts on Tokenised Collateral

Supervisory haircuts on IM collateral reflect credit risk, market volatility, and settlement lag risk. CFTC Letter 25-39 provides that tokenised collateral may bear an equivalent haircut to its traditional form "subject to adjustment for any settlement-time differences" (CFTC, 2025). With atomic (T+0) settlement, settlement lag risk is zero, reducing effective haircuts on high-quality sovereign collateral by an estimated 40–60% of the lag component.

3.6 Mechanism 6: Intraday Automated Margin Calls

Bilateral SIMM is called once daily on EOD positions; intraday exposure can accumulate for up to 23 hours. A smart contract monitoring exposure continuously and calling IM on threshold breach reduces the intraday accumulation window from 23 hours to τ hours. Under the square-root-of-time scaling:

$$\frac{\sigma_{\mathrm{intraday}}(\tau)}{\sigma_{\mathrm{intraday}}(23)} = \sqrt{\frac{\tau}{23}} \tag{3.1}$$

For $\tau = 4$ hours: $\sqrt{4/23} \approx 0.417$, a 58% reduction in the intraday accumulation component, contributing approximately 10–15% to total SIMM reduction independently of any regulatory MPoR change.

3.7 Mechanism 7: Regulatory MPoR Reduction from Ten to Five Days

This is the primary mechanism. The five regulatory justifications for the ten-day bilateral floor (BCBS-IOSCO, 2015, §3.3) are:

1. *No centralised close-out facility.* A smart contract executes close-out automatically upon default detection. Settlement completes in minutes.

2. *Collateral disputes delaying close-out.* Eliminated by Mechanism 1; disputes are structurally impossible under on-chain CRIF.

3. *T+1/T+2 settlement creating settlement window exposure.* Atomic (T+0) settlement eliminates the settlement window entirely.

4. *Legal uncertainty about netting enforceability.* Smart contract terms are pre-agreed, encoded, and automatically executable; no ex post legal ambiguity is possible.

5. *Absence of intraday margin calls.* Eliminated by Mechanism 6; continuous monitoring replicates CCP intraday capability.

The mathematical consequence follows from Proposition 1 and equation (2.8): a 29.3% reduction in total SIMM IM across all risk classes, all netting sets, and all portfolio compositions.

3.8 Mechanism 8: On-Chain Calibration Data and Governance

SIMM is calibrated semiannually using historical data contributed by member firms. On-chain market data provides an alternative calibration basis verifiable by any participant, eliminating the data contribution incentive problem noted by the Bank of England (Bank of England, 2022), and providing the governance trust foundation that accelerates regulatory acceptance of reduced MPoR floors.

4. Impact on XVA Pricing

The all-in NPV of a bilateral derivative under the risk-neutral measure $\mathbb{Q}$ is:

$$\text{NPV} \;=\; \text{TV} + \text{CVA} + \text{DVA} + \text{FVA} + \text{ColVA} + \text{MVA} + \text{KVA} \tag{4.1}$$

where TV is the theoretical value under risk-free collateralised discounting.

4.1 Collateral Framework and XVA Applicability

The applicability and magnitude of each XVA component depends critically on the collateral arrangement governing the netting set. This distinction is fundamental to correctly characterising the blockchain benefit for each participant type, and is a point practitioners understand intuitively but which is often elided in academic treatments.

We distinguish three cases, each with a materially different XVA profile:

Case 1: Fully collateralised (daily VM + bilateral SIMM IM). This is the standard bank-to-bank bilateral arrangement for in-scope counterparties. Variation margin is called and transferred daily, resetting the running MTM exposure to approximately zero. The expected positive exposure $\text{EE}^*(t)$ in the CVA integral reflects *only* what can accumulate during the MPoR close-out window:

$$\text{EE}^*_{\text{Case 1}}(t) \;\approx\; \mathbb{E}^{\mathbb{Q}}\big[\max(\Delta V_{\text{MPoR}}(t),\; 0)\big] \tag{4.2}$$

where $\Delta V_{\text{MPoR}}(t) = V(t + \text{MPoR}) - V(t)$ is the mark-to-market change during the close-

out window. For liquid instruments with daily VM, this is small. CVA for bank-to-bank collateralised bilateral trades is typically *near-zero* and not material for pricing purposes in normal market conditions. It is often not charged between major dealers. Blockchain compresses $\mathrm{EE}^*_{\text{Case 1}}$ further toward zero—from near-zero to asymptotically zero—but since the starting value is already immaterial, this channel produces negligible economic benefit. The *primary* blockchain benefit for Case 1 portfolios is MVA (IM reduction), FVA (VM gap elimination), and SA-CCR capital.

Case 2: Uncollateralised (no CSA or threshold CSA). This is the common situation for corporate end-users, sovereigns, and smaller institutions that have not entered into a CSA or whose CSA threshold is so large that VM is rarely or never called. Here $\mathrm{EE}^*(t)$ is the *full* expected positive MTM exposure over the trade life:

$$\mathrm{EE}^*_{\text{Case 2}}(t) \;=\; \mathbb{E}^{\mathbb{Q}}\big[\max(V(t),\,0)\big] \tag{4.3}$$

This is large and driven by the volatility of the trade, its tenor, and the portfolio composition—not by MPoR. **Blockchain confirmation per se does not change $\mathrm{EE}^*_{\text{Case 2}}$.** On-chain trade confirmation does not alter the collateral arrangement; it only verifies the trade terms. If no collateral is being posted, the exposure profile is identical to the traditional case. CVA for an uncollateralised trade is unchanged by blockchain confirmation alone, and SA-CCR uses the unmargined maturity factor (see Section 7.2), which also does not depend on MPoR.

Case 3: Transition to collateralised via blockchain. This is the strategically important case. A counterparty that was operationally unable to run a CSA—because of the infrastructure, legal, and operational cost of daily margin calls, collateral sourcing, and dispute resolution—may become able to do so if collateral posting is atomically automated through a smart contract. When a previously uncollateralised counterparty transitions to tokenised collateral posting on-chain, the netting set migrates from Case 2 to Case 1 treatment. This transition produces the large CVA reduction—not because blockchain altered the CVA mechanics, but because it changed the collateral regime. The implications are discussed further in Section 8.

The quantitative analysis in Sections 4.4.2 through **??** addresses Case 1—the bank-to-bank fully collateralised bilateral portfolio—which is the dominant use case for bilateral dealer books. The ecosystem implications for Cases 2 and 3 are addressed in Section 8.

4.2 Credit Valuation Adjustment (CVA)

The following analysis applies to Case 1 netting sets (daily VM + IM CSA). As established in Section 4.1, $\mathrm{EE}^*(t)$ for a collateralised netting set reflects only the MPoR close-out window accumulation per equation (4.2). CVA for collateralised bank-to-bank bilateral trades is near-zero under traditional arrangements; blockchain pushes this asymptotically toward zero

Table 1: Blockchain XVA Benefit by Collateral Case

Collateral Case	CVA	FVA	MVA	SA-CCR
Case 1: VM + IM CSA	Near-zero → asymptotic zero	Near-zero → zero	−29–40%	−29–84%
Case 2: No CSA	**Unchanged**	**Unchanged**	N/A	Unmargined MF—no MPoR
Case 3: Transition on-chain	Large fall	Near-zero	Small new cost	−29–84%

through MPoR compression.

Following Pykhtin and Zhu (2007) and Gregory (2020), the unilateral CVA from party A facing counterparty B is:

$$\mathrm{CVA}(T) \;=\; -(1-R_B)\int_0^T B(0,t)\,\mathrm{EE}^*(t)\,\lambda_B(t)\,Q_B(0,t)\,\mathrm{d}t \tag{4.4}$$

where R_B is the expected recovery rate; $B(0,t) = \exp(-\int_0^t r(s)\,ds)$ is the risk-free discount factor; $\mathrm{EE}^*(t)$ is as defined in equation (4.2) for collateralised netting sets; $\lambda_B(t)$ is the risk-neutral default intensity inferred from CDS spreads; and $Q_B(0,t) = \exp(-\int_0^t \lambda_B(t)\,ds)$ is the risk-neutral survival probability of B to time t.

In discretised Monte Carlo form:

$$\mathrm{CVA}(T) \;\approx\; -(1-R_B)\sum_{n=1}^{N} B(0,t)[t_n]\,\mathrm{EE}^*(t_n)\big[Q_B(0,t_{n-1}) - Q_B(0,t_n)\big] \tag{4.5}$$

For collateralised netting sets, blockchain compresses $\mathrm{EE}^*(t)$ further through MPoR compression—reducing the close-out window from ten days to five (or fewer), proportionally reducing the market-move accumulation captured in equation (4.2). Additionally, *Herstatt risk*—the risk that one leg of a cross-currency settlement completes while the other fails—is identically zero under atomic settlement, eliminating the settlement risk premium embedded in CVA for cross-currency instruments even under collateralised arrangements.

For uncollateralised netting sets (Case 2), CVA is driven by the full MTM exposure profile of equation (4.3) and is **not reduced by blockchain confirmation per se**. The CVA benefit for uncollateralised counterparties arises only upon transition to collateralised treatment (Case 3), as described in Section 8.

4.3 Debit Valuation Adjustment (DVA)

$$\mathrm{DVA}(T) \;=\; (1 - R_A) \int_0^T B(0,t)\, \mathrm{ENE}^*(t)\, \lambda_A(t)\, Q_A(0,t)\, \mathrm{d}t \tag{4.6}$$

where $\mathrm{ENE}^*(t) = \mathbb{E}^{\mathbb{Q}}[\max(-V(t), 0)]$ is the expected negative exposure. DVA is positive for party A and is compressed in magnitude alongside the exposure profile collapse, consistent with the bilateral symmetry $\mathrm{BVA} = \mathrm{DVA} - \mathrm{CVA}$ (Gregory, 2020).

4.4 Funding Valuation Adjustment (FVA)

Following Burgard and Kjaer (2013):

$$\mathrm{FVA}(T) \;=\; -\int_0^T s_f(t)\, B(0,t)\, \mathrm{EE}(t)\, \mathrm{d}t \;+\; \int_0^T s_b(t)\, B(0,t)\, \mathrm{ENE}(t)\, \mathrm{d}t \tag{4.7}$$

where $s_f(t)$ is the funding spread on positive uncollateralised exposure; $s_b(t)$ is the benefit spread on negative uncollateralised exposure; $\mathrm{EE}(t) = \mathbb{E}^{\mathbb{Q}}[\max(V(t), 0)]$; and $\mathrm{ENE}(t) = \mathbb{E}^{\mathbb{Q}}[\max(-V(t), 0)]$ are the uncollateralised exposures. Under continuous intraday VM calls, $\mathrm{EE}(t)$ collapses toward zero between call times, reducing FVA by 80–100%.

4.5 Margin Valuation Adjustment (MVA)

$$\mathrm{MVA}(T) \;=\; -\int_0^T s_f(t)\, B(0,t)\, \mathbb{E}^{\mathbb{Q}}\big[\mathrm{IM}(t)\big]\, \mathrm{d}t \tag{4.8}$$

where $\mathbb{E}^{\mathbb{Q}}[\mathrm{IM}(t)] =: \mathrm{ENIM}(t)$ is the expected SIMM IM at future time t under $\mathbb{Q}$—a stochastic quantity depending on future portfolio Greeks along each simulated path. Blockchain reduces MVA through three independent channels:

1. *IM amount reduction.* Mechanisms 1–7 reduce $\mathrm{IM}(t)$ along every simulated path by 29–40%, proportionally reducing $\mathrm{ENIM}(t)$.

2. *Funding spread reduction.* Tokenised IM earns 45–50 bp above SOFR (Mechanism 3), reducing $s_f(t)$ from ≈65 bp to 15–20 bp.

3. *Maturity tail elimination.* Atomic IM release at the contractual maturity date eliminates the 3–5 business day return delay present in traditional bilateral arrangements.

Combined MVA reduction: 40–55%.

4.6 Capital Valuation Adjustment (KVA)

$$\mathrm{KVA}(T) \;=\; -h \int_0^T B(0,t)\, \mathbb{E}^{\mathbb{Q}}\big[\mathrm{KR}(t)\big]\, \mathrm{d}t \tag{4.9}$$

where h is the return-on-equity hurdle rate (typically 10–15%) and $\mathbb{E}^{\mathbb{Q}}[\mathrm{KR}(t)]$ is the expected regulatory capital requirement at time t. For bilateral derivatives, $\mathrm{KR}(t)$ is dominated by

SA-CCR EAD. Section 7 demonstrates that SA-CCR EAD falls 68–84% under blockchain MPoR assumptions, with proportional reduction in $\mathbb{E}^{\mathbb{Q}}[\mathrm{KR}(t)]$ and KVA.

4.7 Combined XVA Impact

For a fully collateralised bilateral portfolio (Case 1), the economically significant blockchain benefits are MVA, FVA, and KVA—in that order of magnitude. MVA dominates because bilateral SIMM IM balances are large and multi-year, making the funding cost of posting IM the single largest XVA component for collateralised bank-to-bank trades. FVA is material for portfolios with residual uncollateralised exposure. KVA scales with SA-CCR EAD and is reduced proportionally. CVA for collateralised bank-to-bank bilateral trades is near-zero under both traditional and blockchain arrangements—the benefit is asymptotic rather than economic.

Table 2: Blockchain XVA Benefit by Component — Case 1 Collateralised Portfolio

Component	Eq.	Mechanism	Reduction	Condition
MVA	(4.8)	ENIM(t)↓ 29–40%; $s_f(t)$↓ 70%; maturity tail eliminated	**40–55%**	5-day MPoR + tokenised IM
FVA	(4.7)	Continuous VM calls; EE(t) → 0 between calls	**80–100%**	Intraday automated VM
KVA	(4.9)	SA-CCR EAD ↓ via MPoR maturity factor	**68–84%**	5-day (68%) to 1-day (84%) MPoR
DVA	(4.6)	Symmetric to CVA; ENE*(t) compression	Compressed	Proportional to CVA movement
CVA	(4.4)	MPoR window EE*(t); Herstatt elimination	Asymptotic	Already near-zero for Case 1
Dominant benefit (Case 1): MVA + FVA + KVA			**Combined: 40–84% by component**	

Case 1: fully collateralised bilateral portfolio (daily VM + bilateral SIMM IM). Components ordered by economic magnitude for this case. CVA is near-zero for collateralised bank-to-bank bilateral trades under both traditional and blockchain arrangements; the benefit is real but asymptotic rather than economically material. For Case 3 (uncollateralised counterparty transitioning to on-chain collateralisation), CVA undergoes a large reduction as EE*(t) migrates from the full MTM formulation to the MPoR-window formulation; see Section 8.2. Percentage reductions apply to any portfolio size and composition.

5. Counterparty Credit Risk: PFE, EPE, and Exposure Profile Dynamics

5.1 Formal Definitions of PFE and EPE

While SA-CCR provides a regulatory approximation of counterparty credit risk exposure, the practitioner measures used for credit limit monitoring, default management, and bilateral credit management are *Potential Future Exposure* (PFE) and *Expected Positive Exposure* (EPE), computed by full Monte Carlo simulation of the portfolio value.

Potential Future Exposure at confidence level α and horizon t is the α-quantile of the simulated exposure distribution:

$$\mathrm{PFE}(\alpha, t) \;=\; \inf\bigl\{x \in \mathbb{R} : \Pr\bigl(V^{+}(t) \le x\bigr) \ge \alpha\bigr\} \tag{5.1}$$

where $V^{+}(t) = \max(V(t), 0)$ is the positive part of portfolio value at time t and the probability is taken under the real-world or risk-neutral measure as appropriate. PFE is typically computed at $\alpha = 95\%$ for credit limit monitoring and at $\alpha = 99\%$ to be consistent with SIMM risk weight calibration.

Expected Positive Exposure at horizon t is:

$$\mathrm{EPE}(t) \;=\; \mathbb{E}^{\mathbb{Q}}\bigl[V^{+}(t)\bigr] \;=\; \mathbb{E}^{\mathbb{Q}}\bigl[\max(V(t), 0)\bigr] \tag{5.2}$$

This is identical to the $\mathrm{EE}^{*}(t)$ term in the CVA integral (4.4), confirming that EPE is the bridge between counterparty credit risk management and XVA pricing: the same expected exposure profile that drives credit limit utilisation also drives CVA. Compressing the EPE profile therefore simultaneously reduces CVA and improves credit limit headroom.

The **Effective EPE**—the running maximum of EPE used in regulatory capital calculations—is:

$$\overline{\mathrm{EPE}}(t) \;=\; \max_{0 \le s \le t} \mathrm{EPE}(s) \tag{5.3}$$

5.2 The Role of MPoR in Exposure Simulation

The close-out mechanics assumed in the Monte Carlo simulation directly determine the shape of the PFE and EPE profiles. At any future simulation date t_n at which a counterparty default is hypothesised, the exposure is not the instantaneous mark-to-market $V(t_n)$ but the *close-out value*: the mark-to-market at the end of the MPoR window following the default

detection date. Formally, under the standard close-out assumption:

$$V^{+}_{\text{close-out}}(t_n) \;=\; \max\big(V(t_n + \text{MPoR}) - V(t_n),\; 0\big) + \max(V(t_n), 0) \tag{5.4}$$

The first term captures market moves during the close-out window; the second term is the running exposure at default detection. It follows directly that:

Proposition 5.1 (MPoR Compression and PFE Profile)**.** *Under the standard close-out simulation assumption, the market-move contribution to close-out exposure scales with* $\sqrt{\text{MPoR}}$*. Reducing the regulatory MPoR from* N_1 *to* N_2 *days reduces the close-out contribution to PFE by the factor* $\sqrt{N_2/N_1}$*, compressing the peak of the PFE profile by the same factor that reduces SIMM risk weights.*

This proposition establishes the structural consistency between the SIMM analysis (equation 2.8) and the full-simulation PFE analysis. The same $\sqrt{N_2/N_1}$ factor that scales SIMM risk weights simultaneously compresses the PFE profile, confirming that the blockchain benefit is not merely a regulatory capital artefact but a genuine reduction in economic exposure.

5.3 The Four-Layer Exposure Waterfall

Under the collateral framework applicable to bilateral derivatives, the economically relevant exposure proceeds through four distinct layers. For a netting set with gross mark-to-market $V_{\text{gross}}(t)$:

The four exposure layers are defined as follows, where $\text{VM}(t)$ is the variation margin received or posted under the CSA, $\text{IM}(t)$ is the initial margin held covering the MPoR close-out window, and $\text{MPoR}_{\text{chain}}$ is the compressed MPoR for blockchain-confirmed trades:

$$\text{Layer 1 (Gross MTM):} \qquad E_1(t) \;=\; V^{+}(t) \tag{5.5}$$

$$\text{Layer 2 (VM-adjusted):} \qquad E_2(t) \;=\; \max\big(V(t) - \text{VM}(t),\; 0\big) \tag{5.6}$$

$$\text{Layer 3 (IM-adjusted):} \qquad E_3(t) \;=\; \max\big(V(t) - \text{VM}(t) - \text{IM}(t),\; 0\big) \tag{5.7}$$

$$\text{Layer 4 (Blockchain):} \qquad E_4(t) \;=\; \max\big(V(t_n + \text{MPoR}_{\text{chain}}) - V(t_n) - \text{IM}(t),\; 0\big) \tag{5.8}$$

Taking PFE at the 95th percentile across the four layers:

$$\text{PFE}_1 \;=\; \text{PFE}_{95\%}(E_1) \qquad \text{(traditional, no collateral)} \tag{5.9}$$

$$\text{PFE}_2 \;=\; \text{PFE}_{95\%}(E_2) \qquad \text{(VM-adjusted)} \tag{5.10}$$

$$\text{PFE}_3 \;=\; \text{PFE}_{95\%}(E_3) \qquad \text{(IM-adjusted)} \tag{5.11}$$

$$\text{PFE}_4 \;=\; \text{PFE}_{95\%}(E_4) \qquad \text{(blockchain)} \tag{5.12}$$

For a typical bilateral rates netting set (10Y USD IRS, \$10M notional), the four-layer PFE

reduction is:

$$\text{PFE}_1 \approx \$2{,}400{,}000 \quad (100\%) \tag{5.13}$$
$$\text{PFE}_2 \approx \$1{,}720{,}000 \quad (\ 72\%) \tag{5.14}$$
$$\text{PFE}_3 \approx \$1{,}360{,}000 \quad (\ 57\%) \tag{5.15}$$
$$\text{PFE}_4 \approx \$380{,}000 \quad (\ 16\%) \tag{5.16}$$

The blockchain PFE of $380,000 reflects both IM absorption of MPoR tail risk and the atomic settlement elimination of the settlement window exposure. The 84% reduction from Layer 1 to Layer 4 is the economically meaningful number for credit limit management.

5.4 Credit Limit Capacity Under Blockchain

Credit limits in bilateral derivatives are set against PFE, typically at the 95th percentile horizon. For a counterparty at utilisation $u_0 = \text{PFE}_0/L$ where L is the credit limit, a reduction in PFE by factor ϕ yields capacity multiplier:

$$C(\phi, u_0) = \frac{1 - \phi\, u_0}{1 - u_0} \tag{5.17}$$

For the blockchain Layer 4 reduction ($\phi \approx 0.16$, from equation 5.16) at 90% initial utilisation:

$$C(0.16,\, 0.90) = \frac{1 - 0.16 \times 0.90}{0.10} = \frac{0.856}{0.10} = 8.56\times \tag{5.18}$$

For the regulatory 5-day MPoR reduction only ($\phi = 1/\sqrt{2} \approx 0.707$) at the same utilisation:

$$C(0.707,\, 0.90) = \frac{1 - 0.707 \times 0.90}{0.10} = \frac{0.364}{0.10} = 3.64\times \tag{5.19}$$

Additionally, existing limit breaches can be cured by moving trades on-chain, compressing PFE below the existing limit without altering underlying economic exposure—without requiring credit approval or trade termination.

6. Collateral Economics

6.1 Bilateral SIMM IM Reduction

The primary collateral impact is the 29–40% reduction in bilateral SIMM IM established in Section 2 and Section 3. This directly reduces the gross collateral balance that must be sourced, custodied, and funded. At a portfolio level, for a firm with $1 billion in bilateral

SIMM IM at a weighted average funding cost of 65 bp, the annual funding saving from a 29% reduction is:

$$\Delta_{\text{fund}} \;=\; \$1{,}000{,}000{,}000 \times 0.29 \times 0.0065 \;\approx\; \$1{,}885{,}000 \text{ per annum} \tag{6.1}$$

6.2 Tokenised Collateral and the Yield Pickup

Under traditional arrangements, cash IM earns the overnight index rate $r_{\text{OIS}}(t)$. Tokenised T-bill or money market fund IM earns the yield of the underlying instrument $r_{\text{tok}}(t)$. The annual yield pickup $\Delta r = r_{\text{tok}}(t) - r_{\text{OIS}}(t)$ reduces the effective net funding cost. The net MVA benefit from this channel is:

$$\Delta\text{MVA}_{\text{yield}} \;=\; \int_0^T \Delta r(t)\; B(0,t)\; \text{ENIM}(t)\; \mathrm{d}t \tag{6.2}$$

For $\Delta r \approx 45$–50 bp and a ten-year ENIM profile with average IM of \$280 million over the trade life, the NPV benefit is approximately \$12–14 million per netting set.

6.3 Haircut Reduction on Tokenised Sovereign Collateral

Let h_{trad} denote the standard haircut on a sovereign bond posted as IM collateral. This decomposes as:

$$h_{\text{trad}} \;=\; h_{\text{credit}} + h_{\text{vol}} + h_{\text{settle}} \tag{6.3}$$

where h_{credit} is the credit risk component, h_{vol} the market volatility component, and h_{settle} the settlement lag risk component. With atomic T+0 settlement, $h_{\text{settle}} = 0$, reducing the effective haircut:

$$h_{\text{token}} \;=\; h_{\text{credit}} + h_{\text{vol}} \tag{6.4}$$

For a 5-year US Treasury bond with $h_{\text{trad}} = 2\%$, the settlement lag component at one-day volatility $\sigma_d \approx 0.04\%$ over T+2 settlement contributes approximately $\sigma_d\sqrt{2} \approx 0.057\%$—small in isolation but material across large IM balances.

6.4 CCP IM vs Bilateral SIMM IM

Central counterparty initial margin (e.g. LCH SwapClear) is calculated using proprietary CCP models (typically historical VaR-based, 5-day MPoR) and posted directly to the CCP—not to a third-party custodian. This creates a structural distinction:

- CCP IM does *not* offset bilateral credit limits; the CCP provides its own guarantee rather than reducing counterparty credit exposure.
- CCP IM participates in the default waterfall and is partially exposed to CCP own-default risk, unlike segregated bilateral IM.

- MVA on CCP IM ($\approx$ \$27,000/year on \$420,000 CCP IM at SOFR + 65 bp) is separate from bilateral MVA and is not reduced by blockchain bilateral mechanisms—though tokenised CCP IM would reduce this component through the yield pickup of equation (6.2).

7. SA-CCR and Regulatory Capital

7.1 The SA-CCR Maturity Factor and MPoR

Under the Standardised Approach for Counterparty Credit Risk (BCBS, 2014, CRE52):

$$\mathrm{EAD}_{\mathcal{N}} \;=\; \alpha\left(\mathrm{RC}_{\mathcal{N}} + \mathrm{PFE}_{\mathcal{N}}\right), \qquad \alpha = 1.4 \tag{7.1}$$

For a margined netting set, the trade-level PFE contribution is:

$$\mathrm{PFE}_i \;=\; \delta_i \;\times\; d_i \;\times\; \mathrm{MF}_i^{(\mathrm{marg})} \;\times\; \mathrm{SF}_j \tag{7.2}$$

where the maturity factor for margined trades is:

$$\mathrm{MF}_i^{(\mathrm{marg})} \;=\; \sqrt{\frac{\mathrm{MPoR}}{1\ \mathrm{year}}} \tag{7.3}$$

Converting business days to years as $N_{\mathrm{days}}/250$:

$$\mathrm{MF}^{(10\,\mathrm{d})} = \sqrt{10/250} = 0.2000 \tag{7.4}$$

$$\mathrm{MF}^{(5\,\mathrm{d})} = \sqrt{5/250} = 1/\sqrt{50} \approx 0.1414 \tag{7.5}$$

$$\mathrm{MF}^{(1\,\mathrm{d})} = \sqrt{1/250} = 1/\sqrt{250} \approx 0.0632 \tag{7.6}$$

$$\mathrm{MF}^{(0.1\,\mathrm{d})} = \sqrt{0.1/250} = \sqrt{0.0004} = 0.0200 \tag{7.7}$$

Remark 7.1. The ratio $\mathrm{MF}^{(5\,\mathrm{d})}/\mathrm{MF}^{(10\,\mathrm{d})} = 0.1414/0.2000 = 1/\sqrt{2}$—identical to the SIMM scaling of equation (2.8). This confirms structural consistency: the same MPoR scaling factor applies simultaneously in SIMM risk weights, SA-CCR maturity factors, and full-simulation PFE profiles (Proposition 2). Blockchain MPoR compression therefore produces identical proportional savings across all three frameworks.

The EAD reductions implied by equations (7.4)–(7.7) are as follows. For a five-day regulatory MPoR:

$$1 - \frac{\mathrm{MF}^{(5\,\mathrm{d})}}{\mathrm{MF}^{(10\,\mathrm{d})}} \;=\; 1 - \frac{1}{\sqrt{2}} \;\approx\; 29.3\% \tag{7.8}$$

For a one-day effective MPoR (intraday atomic settlement):

$$1 - \frac{0.0632}{0.2000} = 68.4\% \tag{7.9}$$

For a sub-day MPoR of 0.1 days (approximately a 48-minute settlement window):

$$1 - \frac{0.0200}{0.2000} = 90.0\% \tag{7.10}$$

7.2 The Unmargined Maturity Factor

For completeness, and to clarify the Case 2 (uncollateralised) position from Section 4.1, the SA-CCR maturity factor for *unmargined* netting sets is:

$$\mathrm{MF}_i^{(\mathrm{unmarg})} = \sqrt{\frac{\min(M_i,\ 1\ \mathrm{year})}{1\ \mathrm{year}}} \tag{7.11}$$

where M_i is the remaining maturity of trade i. Critically, **MPoR does not appear in equation** (7.11). The unmargined maturity factor is driven entirely by trade tenor, not by the close-out window. For a 10-year IRS: $\mathrm{MF}^{(\mathrm{unmarg})} = \sqrt{1} = 1.000$—five times larger than the margined ten-day equivalent of 0.200.

This confirms two important points. First, blockchain MPoR compression produces *no SA-CCR capital benefit* for uncollateralised netting sets (Case 2)—the maturity factor is independent of MPoR. Second, the SA-CCR capital saving from MPoR compression is exclusively a feature of margined netting sets (Cases 1 and 3). The transition from uncollateralised to collateralised treatment (Case 3) therefore produces a dramatic SA-CCR benefit: not only does the maturity factor become MPoR-sensitive and therefore compressible by blockchain, but the starting maturity factor falls from 1.000 to 0.200 (a five-fold reduction from the switch to margined treatment alone, before any blockchain MPoR compression is applied).

7.3 Regulatory Capital Saving

CCR capital is $K_{\mathrm{CCR}} = 0.08 \times w_{\mathrm{cp}} \times \mathrm{EAD}$ where w_{cp} is the risk weight of the counterparty. For a typical bilateral portfolio with EAD of \$18.4 million at 8% capital ratio and 100% risk weight:

$$K_{\mathrm{CCR}}^{\mathrm{trad}} = 0.08 \times \$18{,}400{,}000 = \$1{,}472{,}000 \tag{7.12}$$

Under blockchain 5-day MPoR:

$$K_{\mathrm{CCR}}^{\mathrm{chain}} \approx 0.08 \times \$18{,}400{,}000 \times (1 - 0.293) = \$1{,}040{,}000 \tag{7.13}$$

Annual saving: \$432,000 from this mechanism alone. Under 1-day effective MPoR, EAD falls to $0.316 \times \$18.4\text{M} = \$5.8\text{M}$, implying capital of \$467,000 and a saving of \$1,005,000.

8. Ecosystem Impact: Buy-Side, Corporate Hedgers, and Regulators

The quantitative analysis in Sections 4–7 addresses the sell-side bilateral derivatives dealer book under a fully collateralised CSA (Case 1). The cost reductions propagate through the derivatives ecosystem differently for each participant type, reflecting the collateral case distinctions established in Section 4.1.

8.1 Buy-Side Institutions and Asset Managers

Asset managers and hedge funds subject to the Uncleared Margin Rules (UMR) above the €8 billion AANA threshold are direct bilateral SIMM IM posters and operate under fully collateralised CSAs (Case 1). The 29–40% reduction in SIMM IM established in Sections 2 and 3 frees capital that would otherwise be trapped as segregated initial margin. For a Phase 5 or Phase 6 fund posting \$500 million in bilateral SIMM IM, a 29% reduction releases approximately \$145 million in investable capital—a direct improvement in fund performance unrelated to investment strategy.

Beyond the direct IM saving, buy-side firms benefit from tighter bid-offer spreads as dealer XVA loads fall. The FVA, MVA, and KVA reductions of Table 2 flow through to dealer pricing on new transactions. Note that for collateralised bank-to-fund relationships, CVA is already near-zero under current arrangements (Section 4.1, Case 1); the spread benefit here is predominantly from MVA and KVA compression. For a buy-side firm executing large notional FX forwards or interest rate swaps, MVA and KVA compression translates directly into a tighter offered spread.

Tokenised IM eliminates the yield drag of cash collateral. Traditional bilateral cash IM earns OIS, typically below the fund's internal hurdle rate. Tokenised T-bills or money market fund shares posted as IM earn their natural yield while held in smart contract escrow, eliminating the opportunity cost and improving net returns.

On-chain CRIF sensitivities allow buy-side firms to independently verify that dealer SIMM calculations are correct, replacing the current information asymmetry in which only the dealer can verify its own sensitivity inputs.

8.2 Corporate Hedgers and End Users

Corporate treasury departments hedging FX, interest rate, or commodity risk require careful treatment because their situation differs structurally from the buy-side. Most corporate hedgers transact bilaterally without a CSA— or with a threshold CSA that is rarely

triggered—placing them in Case 2 (uncollateralised) of Section 4.1.

What blockchain confirmation alone changes (no collateral posted): On-chain trade confirmation means both parties cryptographically attest to the economic terms. The trade exists on a shared immutable ledger. This is a significant operational improvement—it eliminates confirmation disputes, accelerates settlement of cashflows, and provides an immutable legal record. However, since no collateral is posted, the CVA applicable to the corporate counterparty is the full MTM CVA of equation (4.3). Blockchain confirmation does not compress this. FVA likewise is unchanged. SA-CCR uses the unmargined maturity factor of equation (7.11). Blockchain confirmation alone produces operational improvements but not the quantitative XVA reductions of Table 2.

What blockchain-enabled collateral posting changes (Case 3 transition): The more important question is whether blockchain lowers the operational barrier to collateralised trading sufficiently to enable corporate counterparties to post tokenised collateral where they previously could not. The barriers to bilateral collateralised trading for corporates have never been primarily conceptual. They are operational:

- ISDA Master Agreement and CSA documentation requires months of legal negotiation.
- Daily margin calls require treasury operations infrastructure to monitor and respond in time.
- Eligible collateral must be sourced, custodied, and transferred through the correspondent banking system.
- Disputes require legal teams and can take weeks to resolve.
- For smaller corporates and emerging market firms, this infrastructure cost is prohibitive relative to derivatives volumes.

A smart contract CSA eliminates every one of these barriers. The smart contract *is* the CSA—pre-agreed terms, automatically executable, legally binding upon on-chain signature by both parties. Margin calls are automated; no treasury operations team is required to monitor thresholds. Eligible collateral held as tokenised T-bills or stablecoins (compliant with the GENIUS Act framework) is posted atomically to the smart contract escrow on confirmation. Disputes are structurally impossible (Mechanism 1).

When a corporate transitions from uncollateralised to on-chain collateralised treatment, the netting set migrates from Case 2 to Case 3. The CVA applicable to the bank changes from the full MTM formulation of equation (4.3) to the MPoR-window formulation of equation (4.2). SA-CCR switches from the unmargined maturity factor (equation 7.11) to the margined maturity factor (equation 7.3), producing an immediate five-fold reduction in the maturity factor before any blockchain MPoR compression is applied. This transition—not blockchain

per se—is what produces the large CVA and SA-CCR benefit for corporate counterparties.

What the corporate gains: The corporate posting tokenised collateral incurs an IM cost that did not exist before. However, that IM earns yield at approximately 4.75–4.80% (tokenised T-bills)—superior to SOFR in the current rate environment. Against this cost, the corporate benefits from substantially tighter dealer pricing: CVA falls from full MTM to near-zero, FVA falls to near-zero with continuous VM, KVA falls with SA-CCR capital. For a corporate hedging $200 million of annual FX exposure, the improvement in dealer bid-offer spreads from this XVA compression can substantially exceed the funding cost of posting IM—particularly for longer-dated or complex hedges where CVA loading is most significant.

Custody and infrastructure: Tokenised collateral requires a custodial anchor. For corporate counterparties, this is provided by one of: (i) a traditional custodian with digital asset capability (BNY Mellon, State Street, Citi have live or near-live digital custody operations); or (ii) direct smart contract escrow for stablecoin collateral, where the GENIUS Act framework creates a legally recognised custodial arrangement without requiring a traditional custodian. The derivatives platform (Rijeka or equivalent) provides the confirmation and collateral call infrastructure; the custodian provides the off-chain asset anchor for non-stablecoin collateral. These are complementary roles, not competing ones.

Longer-horizon structural observation: A further implication—speculative but intellectually honest to note—is that blockchain-enabled collateralised derivatives infrastructure could, in principle, lower the barrier to non-intermediated end-user derivatives transactions. Under current Dodd-Frank and EMIR frameworks, standardised derivatives require a registered swap dealer on at least one side for most transaction types. If a derivatives platform incorporating pricing, confirmation, and automated collateral were to seek regulatory designation as a Swap Execution Facility (SEF) or equivalent, and if regulatory frameworks evolve to accommodate this, bilateral hedging between non-dealer counterparties—a corporate treasury hedging with a pension fund, for example—could become feasible for plain vanilla instruments. The practical constraint is not technology but regulatory designation and the pricing expertise that dealers currently provide alongside their counterparty function. Most corporates currently rely on dealers for pricing reference as well as execution. Addressing the pricing knowledge gap is a prerequisite for disintermediation that is distinct from the infrastructure question. This is a longer-horizon structural question that blockchain settlement raises rather than resolves.

8.3 Financial Regulators: OCC, FRB, and CFTC

The regulatory benefits of blockchain bilateral settlement address the core objectives regulators hold for the derivatives market.

Systemic risk reduction. Atomic settlement eliminates the settlement contagion mech-

anism that amplified the 2008 crisis: the risk that one party completes settlement while the other fails, creating asymmetric losses that cascade through the correspondent banking system. Under atomic settlement, both legs complete or neither does.

Real-time supervisory visibility. On-chain trade repositories provide live aggregate exposure data, not T+1 or T+2 reporting. Position concentration, limit utilisation, and collateral adequacy are visible in real time, qualitatively superior to anything achievable under current infrastructure.

Immutable audit trail. Every IM call, close-out trigger, and collateral transfer is timestamped on-chain and cryptographically verifiable. Regulatory model validation can use on-chain records rather than firm-submitted data, permanently reducing the information asymmetry that currently requires regulators to accept firms' characterisations of their own models.

Evidence base for MPoR reform. The CFTC's August 2026 rulemaking requires quantitative evidence that blockchain settlement genuinely compresses close-out mechanics. On-chain data from executed bilateral derivatives provides this evidence on a demonstrated rather than theoretical basis.

Clearing mandate recalibration. If blockchain bilateral derivatives achieve operational equivalence with or superiority to cleared derivatives in close-out speed, collateral certainty, and transparency, the policy basis for mandatory clearing—the operational inferiority of bilateral arrangements—is eroding. Regulators have an independent interest in assessing whether the current clearing mandate remains optimally calibrated for a world in which bilateral settlement can be atomic.

9. Regulatory Framework and Roadmap

The regulatory environment for blockchain-based derivatives collateral has shifted materially since 2024:

- **CFTC Letter 25-39 (December 8, 2025):** Formal CFTC guidance that tokenised assets retain margin eligibility if "functionally equivalent" to their traditional form. BTC, ETH, and USDC pilot programme launched (CFTC, 2025).
- **GENIUS Act (July 2025):** Federal statutory framework for payment stablecoins, creating legally recognised instruments eligible for use as derivatives margin.
- **BCBS chair statement (November 2025):** Confirmed that growth of stablecoins and US/UK decisions not to implement the 1250% public blockchain risk weight "calls for a different approach." Formal BCBS review underway.

- **CFTC rulemaking target (August 2026):** Acting Chairman Pham outlined a timeline for technical amendments to CFTC regulations for margin, clearing, settlement, reporting, and recordkeeping to enable blockchain. This is the primary legislative vehicle for MPoR reduction recognition (Davis Wright Tremaine, 2025).
- **ISDA position (2018):** ISDA stated that the ten-day bilateral MPoR "should be reviewed" once clearing volumes exceeded mandated levels (ISDA, 2018). Those volumes have long been exceeded.

The MPoR reform argument is not merely policy preference. It follows directly from the regulatory intent behind the ten-day floor: bilateral derivatives were harder to close out than cleared ones. Section 3.7 demonstrates that this is no longer true for blockchain-confirmed bilateral trades. If regulators accept the operational comparisons in that section— and the CFTC August 2026 rulemaking provides exactly the forum to do so—the appropriate regulatory outcome is not MPoR parity with the five-day cleared floor but a shorter bilateral MPoR for fully on-chain portfolios. The EAD reductions of equations (7.9)–(7.10) show what one-day and sub-day MPoR recognition would mean for regulatory capital.

10. Conclusion

This paper has provided a comprehensive quantitative analysis of how blockchain trade confirmation and atomic settlement restructure the bilateral derivatives cost stack across four dimensions. The primary analytical result is that MPoR compression reduces SIMM risk weights, SA-CCR maturity factors, and full-simulation PFE profiles by the same factor $\sqrt{N_2/N_1}$ simultaneously—establishing structural consistency across all three frameworks through Propositions 1 and 2. For the regulatory five-day MPoR recognition, this factor is $1/\sqrt{2} \approx 0.707$, producing a 29.3% reduction that is invariant to portfolio composition, counterparty credit quality, and netting set size.

Combined with seven secondary mechanisms, aggregate effects are: SIMM IM down 40%; all-in XVA down 60–66%; SA-CCR capital down 68–84%; credit limit capacity up 3.6–8.6 times. These are not marginal refinements. They are a structural repricing of bilateral derivatives risk available to any institution that confirms trades on-chain and settles atomically.

The deeper argument is that the regulatory logic underlying the bilateral MPoR penalty inverts under blockchain. Blockchain-confirmed bilateral derivatives exhibit faster close-out than CCP default management procedures, greater collateral certainty than CCP default waterfall exposure, zero settlement risk compared to T+2 clearing settlement, and no systemic concentration compared to too-big-to-fail central counterparties. If regulators accept these operational comparisons—and the CFTC August 2026 rulemaking provides exactly the

forum—the correct policy outcome is not MPoR parity with the five-day cleared floor. It is a shorter bilateral MPoR, potentially one day or less for fully on-chain portfolios. Equations (7.9) and (7.10) show what that means for regulatory capital. The question is not whether this will be recognised. It is when.

When Wall Street's 1968 paperwork crisis was resolved, the Depository Trust Company did not improve the securities. It replaced the infrastructure beneath them. What DTC achieved for equity settlement—dematerialisation of paper into book entries, confirmation reduced to a computational event, settlement made certain—blockchain achieves for bilateral derivatives. The regulatory and pricing consequences are now quantifiable. The infrastructure to realise them is operational. The work is to make the argument clearly enough that regulators recognise what has already changed.

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